

SQL Fundamentals

i) Selecting and Sorting

① Retrieve all columns from employee table

```
SELECT id,  
       first-name,  
       last-name,  
       department, OR  
       salary,  
       hire-date,  
       city  
FROM   Table employees ;
```

Id	First-name	last-name	department	Salary	hire-date	city
1	Alice	Green	IT	70000	2020-01-10	Johannesburg
2	Brian	Lee	HR	45000	2019-03-22	Cape Town
3	Cathy	Zulu	Finance	65000	2018-07-18	Durban
4	David	Mokoena	Marketing	50000	2021-11-05	Pretoria
5	Eva	Naidoo	IT	72000	2017-09-30	Johannesburg

② Find all unique departments

```
SELECT DISTINCT department  
FROM   Table employees ;
```

department
IT
HR
Finance
Marketing

②

③ Retrieve First and last names ordered by Salary descending

```
SELECT First-name,
        last-name
FROM    table-employees
ORDER BY Salary DESC ;
```

First-name	last-name	Salary
Eva	Naidoo	72000
Alice	Green	70000
Cathy	Zulu	65000
David	Mokoena	50000
Brian	Lee	45000

④ Retrieve the top 3 highest paid employees

```
SELECT id,
        Firstname,
        Last name,
        Salary
FROM    table-employees
ORDER BY Salary DESC
LIMIT 3 ;
```

id	First-name	Last-name	Salary
5	Eva	Naidoo	72000
1	Alice	Green	70000
3	Cathy	Zulu	65000

③

5) Find employees in the IT department.

```
SELECT id,
       first-name,
       last-name,
       department
FROM   table - employees
WHERE  department = 'IT' ;
```

id	first-name	last-name	department
1	Alice	Green	IT
5	Eva	Naidoo	IT

6) Find employees in Finance with Salary > 60 000

```
SELECT id,
       first-name,
       last-name,
       department,
       Salary
FROM   table - employees
WHERE  Salary > 60 000 AND Department = Finance
```

id	first-name	last-name	department	Salary
1	Alice	Green	IT	70 000
3	Cathy	Zulu	Finance	65 000
5	Eva	Naidoo	IT	72 000

(A)

7) Find Employees in HR or marketing

```
SELECT id,  
       first-name,  
       last-name,  
       department  
FROM   table-employees  
WHERE  department = 'HR' OR 'marketing';
```

id	first-name	last-name	department
2	Brian	Lee	HR
4	David	Mokoena	marketing

8) Find employees not in IT

```
SELECT id,  
       first-name,  
       last-name,  
       department  
FROM   table-employees  
WHERE  NOT department = 'IT';
```

id	first-name	last-name	department
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
4	David	Mokoena	marketing

9) Find employees in IT, HR, Finance using IN

```
SELECT id,
       first-name,
       last-name,
       department
FROM   table-employees
WHERE  department IN ('IT', 'HR', 'Finance');
```

id	First-name	Last-name	department
1	Alice	Green	IT
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
5	Eva	Maidoo	IT

10) Find employees in IT with salary 765000 and city Johannesburg.

```
SELECT id,
       first-name,
       last-name
WHERE  department = 'IT' AND city = 'Johannesburg'
```

10) Find employees in IT with salary 765000 and city Johannesburg

```
SELECT id,
       first-name,
       last-name,
       department,
```


⑥

FROM table - employees
WHERE department = 'IT' AND SALARY > 65000 AND city = 'Johannesburg';

id	First-name	last-name	department	Salary	City
1	Alice	Green	IT	70 000	Johannesburg
5	Eva	Naidoo	IT	72 000	Johannesburg